

2.8 同余式 (2)

Lemma: p 和 q 是任意的两个素数, 则有:

$$p \neq q \Leftrightarrow \gcd(p, q) = 1 \Leftrightarrow p \nmid q \text{ 且 } q \nmid p$$

Proof: ~~左边 $\Rightarrow$ 中间~~ $\because p$ 和 q 是任意的两个素数 $\therefore p, q \in \mathbb{Z}_{\geq 2}$

$$\because p, q \in \mathbb{Z}_{\geq 2} \quad \therefore \gcd(p, q) \in \mathbb{Z}_{\geq 1}$$

左边 $\Rightarrow$ 中间: $\because p$ 是素数 $\therefore p$ 的全部因数为 $1, -1, p, -p$
 $\because q$ 是素数 $\therefore q$ 的全部因数为 $1, -1, q, -q$

$$\therefore \gcd(p, q) = 1 \quad \therefore \text{左边} \Rightarrow \text{中间} \text{ 得证}$$

中间 $\Rightarrow$ 左边: 假设 $p = q$, 则有:

$$\because p \mid p \text{ 且 } p = q \quad \therefore p \mid q \quad \therefore p \text{ 是 } p \text{ 和 } q \text{ 的一个公因数}$$

$$\because p \in \mathbb{Z}_{\geq 2} \quad \therefore p \in \mathbb{Z}_{\geq 1} \quad \therefore \gcd(p, q) \geq p \geq 2 > 1 \quad \therefore 1 > 1$$

$$\text{矛盾} \quad \therefore p \neq q \quad \therefore \text{中间} \Rightarrow \text{左边} \text{ 得证} \quad \therefore \text{左边} \Leftrightarrow \text{中间}$$

右边 $\Rightarrow$ 右边: 假设 $p \mid q$ 或 $q \mid p$, 则有:

$$\because p \text{ 是素数} \quad \therefore p \text{ 的全部因数为 } 1, -1, p, -p$$

$$\because q \text{ 是素数} \quad \therefore q \text{ 的全部因数为 } 1, -1, q, -q$$

$$\text{如果 } p \mid q, \text{ 则有: } p \text{ 是 } q \text{ 的因数} \quad \therefore p \in \mathbb{Z}_{\geq 2} \quad \therefore p = q \text{ 矛盾}$$

$$\text{如果 } q \mid p, \text{ 则有: } q \text{ 是 } p \text{ 的因数} \quad \therefore q \in \mathbb{Z}_{\geq 2} \quad \therefore q = p \text{ 矛盾}$$

$$\therefore p \nmid q \text{ 且 } q \nmid p \quad \therefore \text{右边} \Rightarrow \text{右边} \text{ 得证}$$

右边 $\Rightarrow$ 左边 假设 $p=q$ 则有:

$$\therefore p|p, p=q \quad \therefore p|q \quad \therefore q|q, p=q \quad \therefore q|p$$

$$\therefore p|q \text{ 且 } q|p \text{ 矛盾} \quad \therefore p \neq q \quad \therefore \text{右边} \Rightarrow \text{左边得证}$$

$$\therefore \text{左边} \Leftrightarrow \text{右边} \quad \square$$

Lemma ($\varphi(n)$ 是积性函数) 对 $\forall m, n \in \mathbb{Z}_{\geq 1}$, $\gcd(m, n) = 1$, 有:

$$\varphi(nm) = \varphi(n) \varphi(m)$$

对 $\forall$ 素数 p ,
Proof: 假设 ~~素数 p s.t.~~ $p|m$ 且 $p|n$, 则有: p 是 m 和 n 的公因数

$$\therefore \gcd(m, n) \geq p \geq 2 > 1 \quad \therefore 1 > 1 \text{ 矛盾} \quad \therefore p \nmid m \text{ 或 } p \nmid n$$

$\therefore$ 对 $\forall$ 素数 p , 有: $p \nmid m$ 或 $p \nmid n$.

$\therefore n \in \mathbb{Z}_{\geq 1}$ $\therefore n$ 有素因子分解: $n = p_1^{a_1} \cdots p_r^{a_r}$, 其中 $r \in \mathbb{Z}_{\geq 0}$ ($r=0$ 对应于 $n=1$ 的情形), $p_1, \dots, p_r$ 是相异素数, $a_1, \dots, a_r \in \mathbb{Z}_{\geq 1}$

$$\therefore p_1|n \quad \therefore p_1 \nmid m \quad \therefore p_2|n \quad \therefore p_2 \nmid m, \dots, \therefore p_r|n \quad \therefore p_r \nmid m$$

$\therefore m \in \mathbb{Z}_{\geq 1}$ $\therefore m$ 有素因子分解: $m = p_{r+1}^{a_{r+1}} \cdots p_t^{a_t}$, 其中 $t \in \mathbb{Z}$ 且 $t \geq r$ ($t=r$ 对应于 $m=1$ 的情形), $p_{r+1}, \dots, p_t$ 是相异素数, $p_1, \dots, p_t$ 是相异素数, $a_{r+1}, \dots, a_t \in \mathbb{Z}_{\geq 1}$

$$\therefore nm = p_1^{a_1} \cdots p_r^{a_r} p_{r+1}^{a_{r+1}} \cdots p_t^{a_t}$$

$$\therefore \varphi(nm) = (nm) \prod_{i=1}^t \left(1 - \frac{1}{p_i}\right) = (nm) \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right) \cdot \prod_{i=r+1}^t \left(1 - \frac{1}{p_i}\right)$$

$$= \left(n \cdot \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right) \right) \cdot \left(m \cdot \prod_{i=r+1}^t \left(1 - \frac{1}{p_i}\right) \right) = \varphi(n) \cdot \varphi(m) \quad \square$$

Remark: 之后给出利用中国剩余定理的证明方法.

Lemma: 对 $\forall x \in [3, +\infty)$, 有: $\sqrt{x} < x-1$

Proof: 令 $f(x) = x-1-\sqrt{x}$, $x \in [0, +\infty)$. 则有:

$$f'(x) = 1 - 0 - \frac{1}{2}x^{-\frac{1}{2}} = 1 - \frac{1}{2\sqrt{x}}, \quad x \in (0, +\infty)$$

$$\therefore f'(x) \text{ 在 } (0, +\infty) \text{ 上严格单调递增, } f'(\frac{1}{4}) = 1 - \frac{1}{2 \times \frac{1}{2}} = 1 - 1 = 0$$

$$\therefore \text{对 } \forall x \in (\frac{1}{4}, +\infty), \text{ 有: } f'(x) > 0$$

$$\therefore f(x) \text{ 在 } (\frac{1}{4}, +\infty) \text{ 上严格单调递增}$$

$$\therefore f(3) = 3-1-\sqrt{3} = 2-\sqrt{3} = \sqrt{4}-\sqrt{3} > 0$$

$$\therefore \text{对 } \forall x \in [3, +\infty), \text{ 有: } f(x) \geq f(3) > 0$$

$$\therefore \text{对 } \forall x \in [3, +\infty), \text{ 有: } \sqrt{x} < x-1 \quad \square$$

Lemma: $\alpha \in \mathbb{Z}_{\geq 2}$, $\beta \in \mathbb{Z}_{\geq 1}$, 则有:

$$\alpha^{\frac{\beta}{2}} \leq \alpha^{\beta-1}(\alpha-1) \Leftrightarrow \alpha \neq 2 \text{ 或 } \beta \neq 1$$

Proof: ($\Rightarrow$): 假设 $\alpha=2$ 且 $\beta=1$. 则有:

$$\alpha^{\frac{\beta}{2}} = 2^{\frac{1}{2}} = \sqrt{2}, \quad \alpha^{\beta-1}(\alpha-1) = 2^0(2-1) = 1 \times 1 = 1$$

$$\therefore \sqrt{2} = \alpha^{\frac{\beta}{2}} \leq \alpha^{\beta-1}(\alpha-1) = 1 < \sqrt{2} \text{ 矛盾.}$$

$$\therefore \alpha \neq 2 \text{ 或 } \beta \neq 1.$$

$$(\Leftarrow): \text{若 } \beta \neq 1, \text{ 则有: } \beta \in \mathbb{Z}_{\geq 2} \quad \therefore \beta \geq 2 \quad \therefore \frac{\beta}{2} \geq 1 \quad \therefore \frac{\beta}{2} - 1 \geq 0$$

$$\therefore \frac{\beta}{2} + \frac{\beta}{2} - 1 \geq \frac{\beta}{2} \quad \therefore \beta - 1 \geq \frac{\beta}{2}$$

指数函数 $y = \alpha^x$ 在 $\alpha \geq 2$ 时严格单调递增.

$$\therefore \alpha^{\beta-1} \geq \alpha^{\frac{\beta}{2}} > 0$$

$$\therefore \alpha \geq 2 \quad \therefore \alpha - 1 \geq 1 \quad \therefore \alpha^{\beta-1} > 0 \quad \therefore (\alpha - 1)\alpha^{\beta-1} \geq 1 \cdot \alpha^{\beta-1}$$

$$\therefore \alpha^{\beta-1}(\alpha - 1) \geq \alpha^{\beta-1} \geq \alpha^{\frac{\beta}{2}}$$

$$\therefore \alpha^{\frac{\beta}{2}} \leq \alpha^{\beta-1}(\alpha - 1)$$

若 $\beta = 1$, 则有: $\alpha \neq 2$. $\therefore \alpha \in \mathbb{Z}_{\geq 2}$ 且 $\alpha \neq 2$ $\therefore \alpha \in \mathbb{Z}_{\geq 3}$. $\therefore \alpha \geq 3$

$$\therefore \sqrt{\alpha} < \alpha - 1 \quad \therefore \alpha^0 = 1 \quad \therefore \alpha^{1-1} = \alpha^0 = 1$$

$$\therefore \alpha^{\frac{1}{2}} < 1 \cdot (\alpha - 1) = \alpha^{1-1}(\alpha - 1)$$

$$\therefore \alpha^{\frac{\beta}{2}} < \alpha^{\beta-1}(\alpha - 1)$$

$$\therefore \alpha^{\frac{\beta}{2}} \leq \alpha^{\beta-1}(\alpha - 1) \quad \square$$

Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, n 是奇数, 有: $\varphi(n) \geq \sqrt{n}$

Proof: 当 $n = 1$ 时, $\varphi(n) = \varphi(1) = 1 \geq 1 = \sqrt{1} = \sqrt{n}$. 结论得证.

当 $n \geq 3$ 时, n 有素因子分解 $n = p_1^{a_1} \cdots p_r^{a_r}$, 其中 $r \in \mathbb{Z}_{\geq 1}$, $p_1, \dots, p_r$ 是相异素数, $a_1, \dots, a_r \in \mathbb{Z}_{\geq 1}$.

$$\therefore \varphi(n) = \prod_{i=1}^r (p_i^{a_i} - p_i^{a_i-1}) = \prod_{i=1}^r (p_i^{a_i-1} \cdot p_i - p_i^{a_i-1} \cdot 1) = \prod_{i=1}^r p_i^{a_i-1} (p_i - 1)$$

对 $\forall i = 1, \dots, r$, $\therefore n$ 是奇数 $\therefore p_i \neq 2 \quad \therefore p_i \in \mathbb{Z}_{\geq 3}$.

$$\therefore p_i \in \mathbb{Z}_{\geq 3}, a_i \in \mathbb{Z}_{\geq 1} \quad \therefore p_i^{a_i-1} (p_i - 1) \geq p_i^{\frac{a_i}{2}} > 0$$

$$\therefore \prod_{i=1}^r p_i^{a_i-1} (p_i - 1) \geq \prod_{i=1}^r p_i^{\frac{a_i}{2}} \quad \therefore \varphi(n) \geq \sqrt{n} \quad \square$$

Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, $4|n$, 有: $\varphi(n) \geq \sqrt{n}$.

proof: $\because n \in \mathbb{Z}_{\geq 1}$ 且 $4|n \quad \therefore n \in \mathbb{Z}_{\geq 4}$

$\therefore n$ 有素因子分解: $n = 2^{a_0} p_1^{a_1} \cdots p_r^{a_r}$, 其中 $r \in \mathbb{Z}_{\geq 0}$ (当 $r=0$ 时, $n=2^{a_0}$).

$p_1, \dots, p_r$ 是 ≥ 3 的相异素数, $a_0 \in \mathbb{Z}_{\geq 2}$, $a_1, \dots, a_r \in \mathbb{Z}_{\geq 1}$.

$$\because a_0 \in \mathbb{Z}_{\geq 2} \quad \therefore a_0 \neq 1 \quad \therefore 2^{a_0-1} \left(\frac{2}{2} - 1 \right) \geq 2^{\frac{a_0}{2}} > 0 \quad \therefore 2^{a_0-1} \geq 2^{\frac{a_0}{2}} > 0$$

$$\text{对 } \forall i=1, \dots, r \quad \therefore p_i \in \mathbb{Z}_{\geq 3}, a_i \in \mathbb{Z}_{\geq 1} \quad \therefore p_i^{a_i-1} (p_i - 1) \geq p_i^{\frac{a_i}{2}} > 0$$

$$\therefore \varphi(n) = (2^{a_0} - 2^{a_0-1}) \prod_{i=1}^r (p_i^{a_i} - p_i^{a_i-1})$$

$$= 2^{a_0-1} (2-1) \prod_{i=1}^r p_i^{a_i-1} (p_i - 1)$$

$$\geq 2^{\frac{a_0}{2}} \prod_{i=1}^r p_i^{\frac{a_i}{2}} = \sqrt{n} \quad \square$$

Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, 有: $\varphi(n) \geq \sqrt{\frac{n}{2}}$

Proof: 对 n 分类讨论如下:

① 若 n 是奇数, 则有: $n \in \mathbb{Z}_{\geq 1}$ 且 n 是奇数. $\therefore n - \frac{n}{2} = \frac{n}{2} > 0 \quad \therefore n > \frac{n}{2} > 0$

$$\therefore \varphi(n) \geq \sqrt{n} > \sqrt{\frac{n}{2}}$$

② 若 n 是偶数 且 $4|n$, 则有: $n \in \mathbb{Z}_{\geq 1}$ 且 $4|n \quad \therefore n - \frac{n}{2} = \frac{n}{2} > 0 \quad \therefore n > \frac{n}{2} > 0$

$$\therefore \varphi(n) \geq \sqrt{n} > \sqrt{\frac{n}{2}}$$

③ 若 n 是偶数 且 $4 \nmid n$, 则有: $\therefore n$ 是偶数 $\therefore 2|n \quad \therefore 4 \nmid n$

~~且有素~~ $n = 2\lambda$, 其中 $\lambda \in \mathbb{Z}_{\geq 1}$, λ 是奇数. $\therefore \gcd(2, \lambda) = 1$

$$\therefore \varphi(n) = \varphi(2\lambda) = \varphi(2) \varphi(\lambda) = 1 \cdot \varphi(\lambda) = \varphi(\lambda) \geq \sqrt{\lambda} = \sqrt{\frac{n}{2}} \quad \square$$

Remark: 关于③再补充一些证明过程:

$\because n \in \mathbb{Z}_{\geq 1}$ 且 n 是偶数 $\therefore n = 2\lambda, \lambda \in \mathbb{Z}_{\geq 1}$.

假设 λ 是偶数, 则有 $2|\lambda \therefore \frac{\lambda}{2} \in \mathbb{Z} \therefore \frac{2\lambda}{4} \in \mathbb{Z} \therefore \frac{n}{4} \in \mathbb{Z}$

$\therefore 4|n$. 这与 $4 \nmid n$ 矛盾. $\therefore \lambda$ 是奇数.

Lemma: $\lim_{n \rightarrow +\infty} \varphi(n) = +\infty$

Proof: $\because$ 对 $\forall n \in \mathbb{Z}_{\geq 1}$, 有: $\varphi(n) \geq \sqrt{\frac{n}{2}}$

又 $\because \lim_{n \rightarrow +\infty} \sqrt{\frac{n}{2}} = +\infty$

$\therefore \lim_{n \rightarrow +\infty} \varphi(n) = +\infty$

